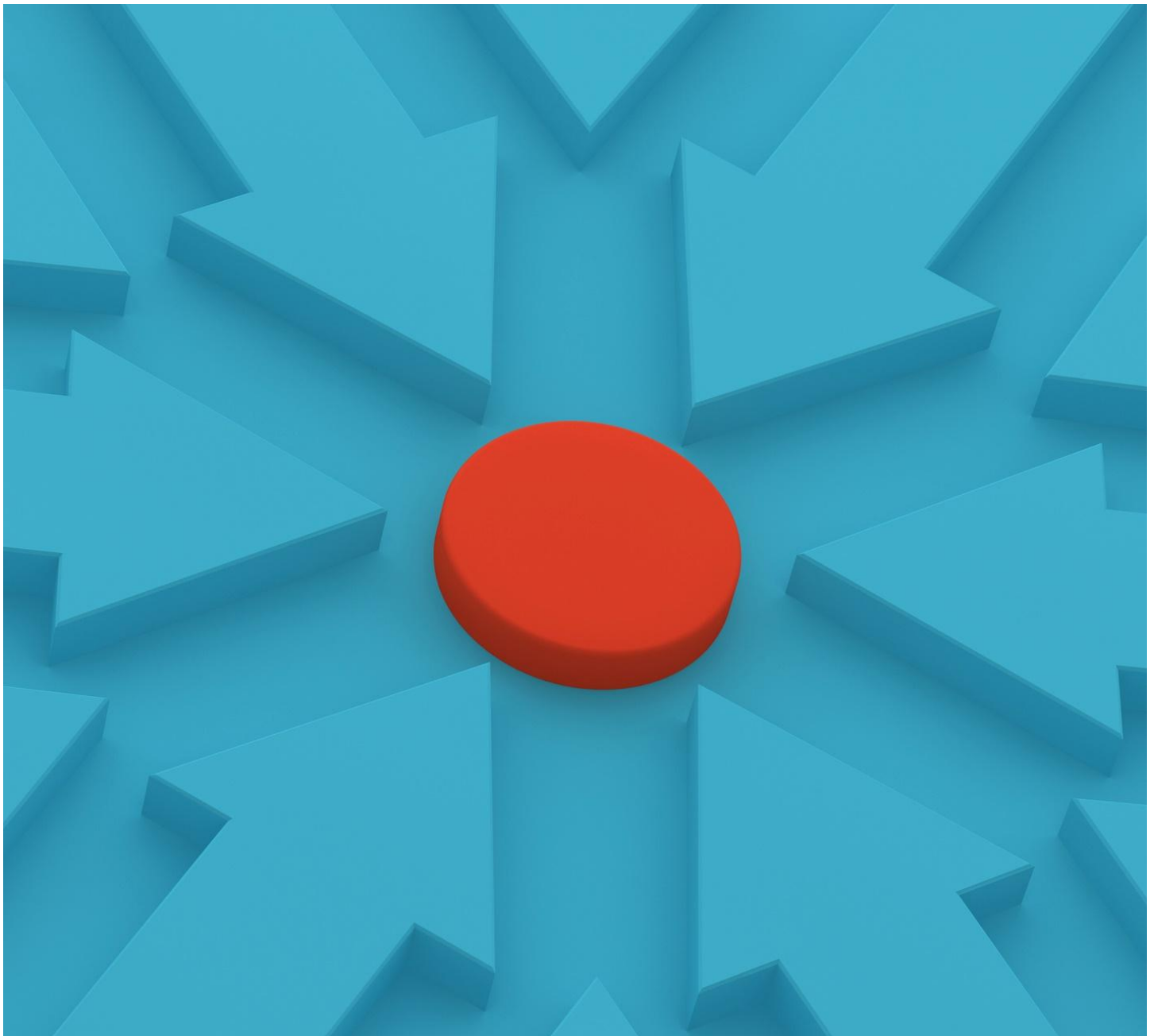




RETAIL SALES ANALYTICS EDA- REPORT

REPORT BY- SANTOSH KR. MAJHI

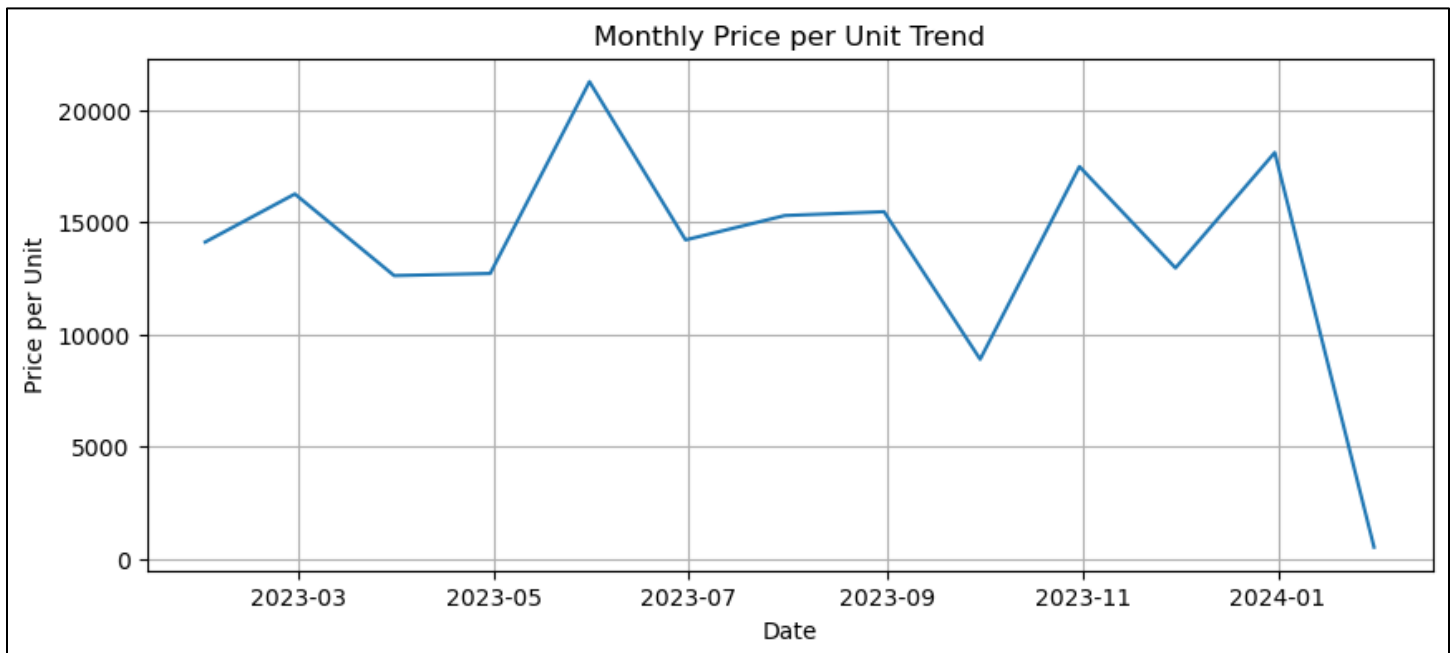
Sales Performance Insights

Insights:

- Monthly revenue fluctuates significantly across the year.
- **May shows the highest sales (53,150)** — a major revenue peak.
- **February (44,060)** and **December (44,690)** also demonstrate strong performance.
- **September has the lowest sales (23,620)**, which may indicate a seasonal slowdown.
- Overall, the store has **consistent sales**, with occasional high-volume months.

Visualizations

Line Chart: Monthly/weekly total sales trend



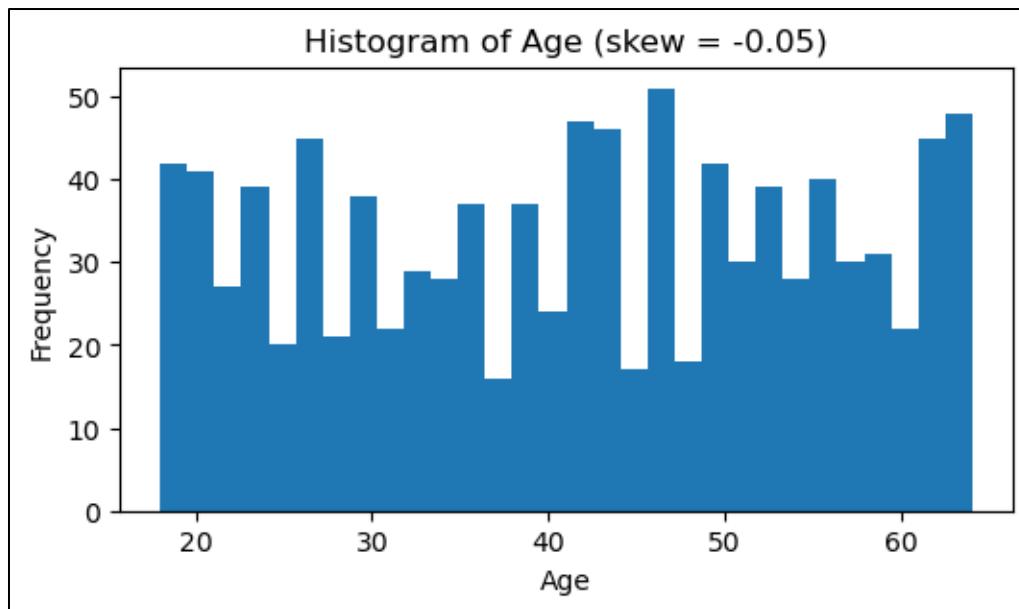
Customer Demographic Insights

Insights:

- Customer age ranges from **18 to 64**, with an average of **41.3 years**.
- Histogram shows most customers fall within the **28–55** range.
- This indicates the business primarily attracts a **mature customer base**.
- No extreme outliers in age distribution.

Visualizations

Histogram: Distribution of Age



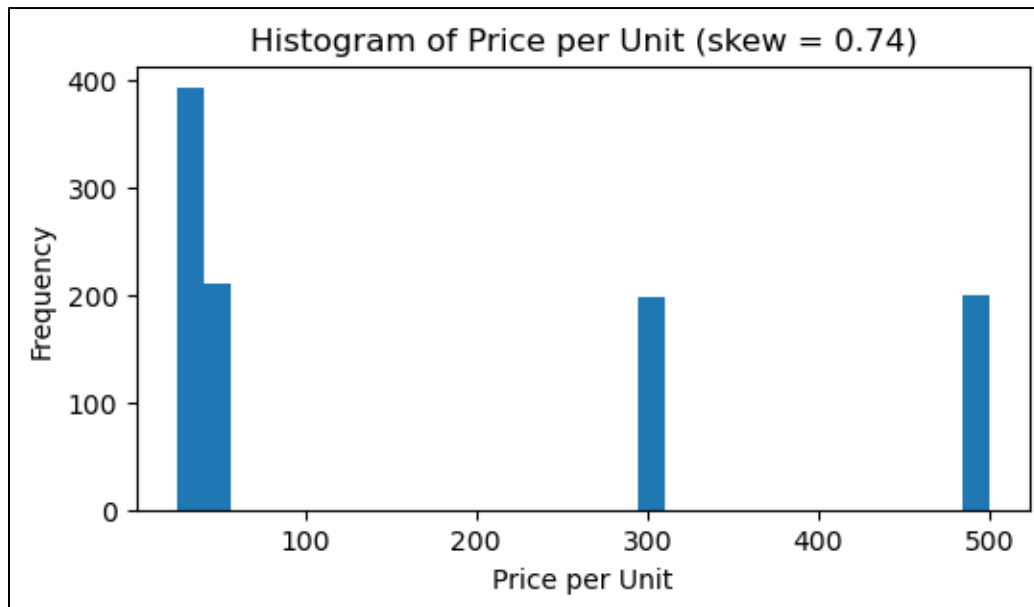
Product Behavior Insights

Insights:

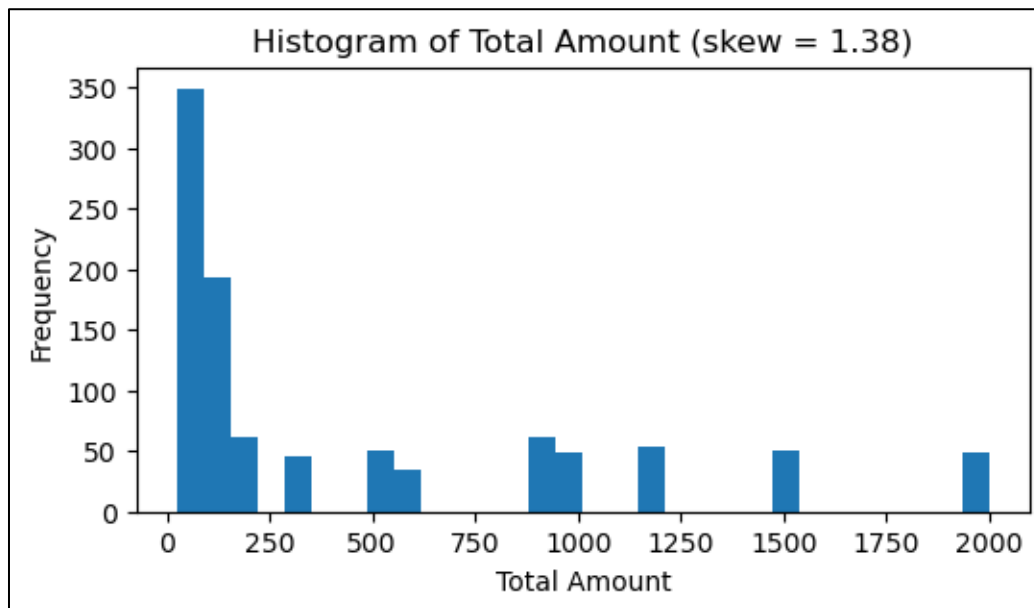
- **Price per Unit** ranges from 30 to 500, showing a wide variety in product pricing.
- **Total Amount** distribution is **right-skewed** — meaning:
 - Most customers make small-to-medium purchases.
 - A few large purchases significantly increase total revenue.
- **Quantity** histogram shows most purchases are in the **lower quantity range**, with occasional bulk purchases.

Visualizations

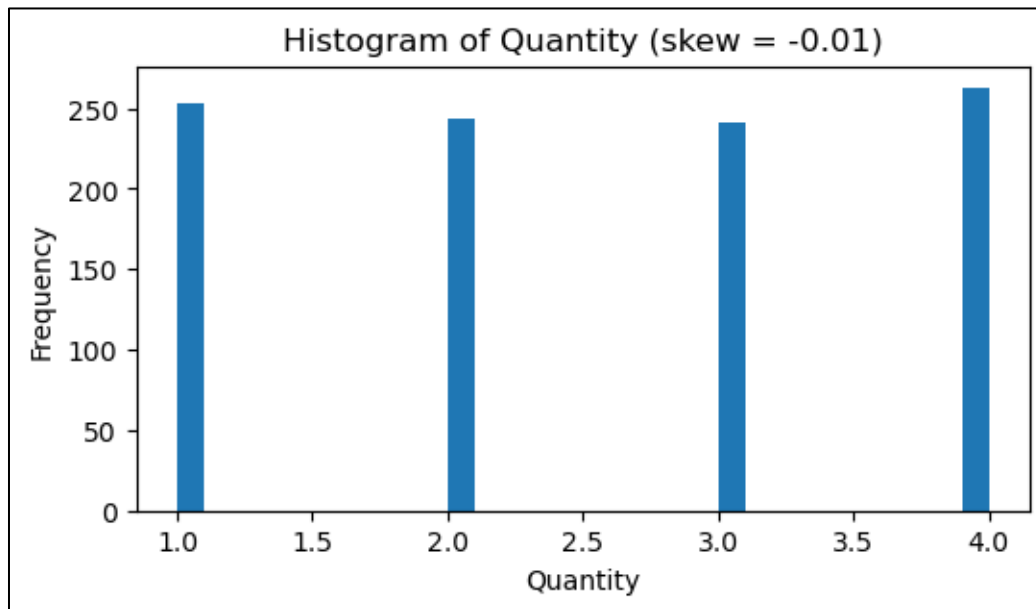
Histogram: Price per Unit distribution



Histogram: Total Amount



Histogram: Quantity



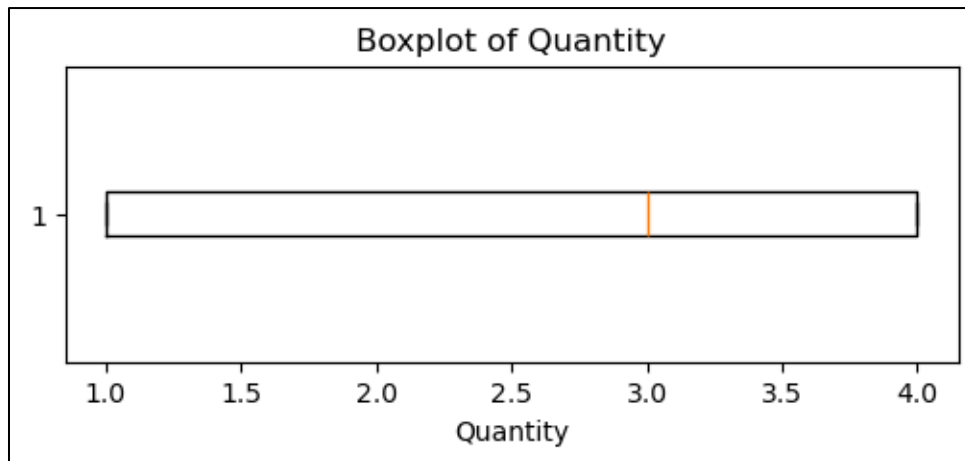
Outlier & Anomaly Detection

Insights:

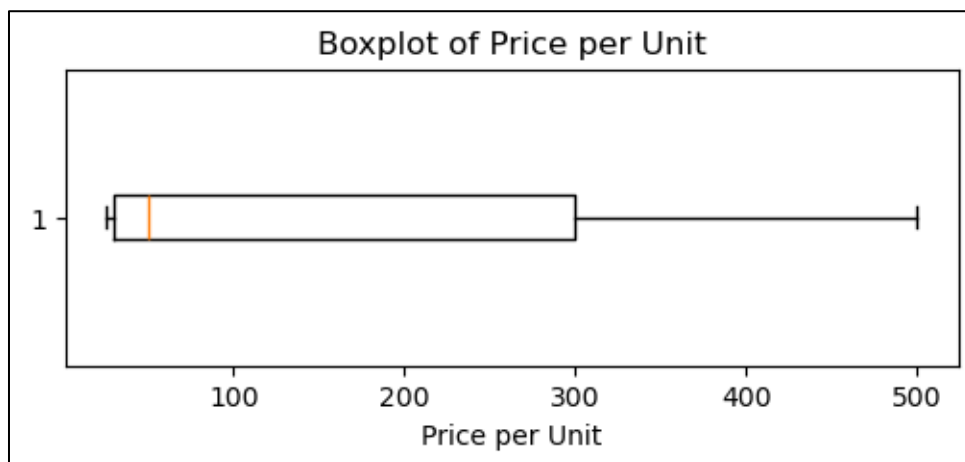
- Boxplots reveal outliers in:
 - **Price per Unit** (very high-priced items)
 - **Total Amount** (high-value transactions)
 - **Quantity** (bulk purchase cases)
- These outliers appear to be **valid business transactions**, not data errors.
- They may represent:
 - Premium products
 - High-end electronics
 - Multi-item orders

Visualizations

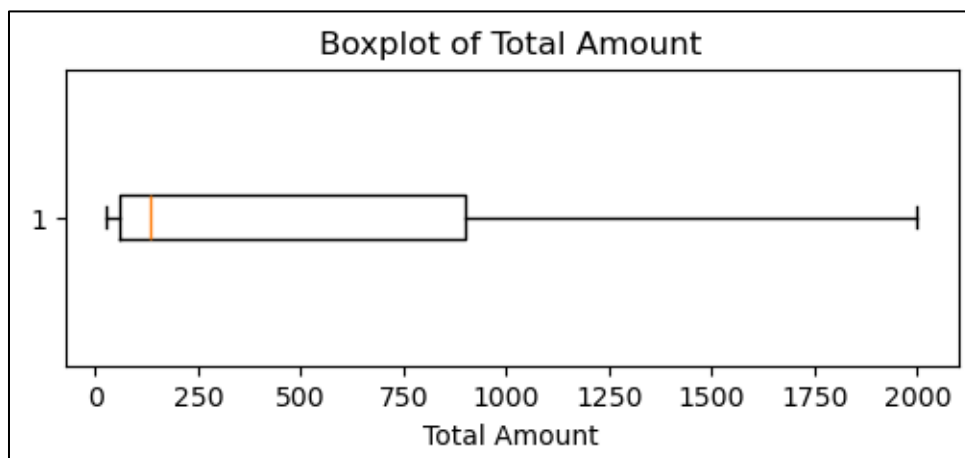
Boxplot: Quantity



Boxplot: Price per Unit



Boxplot: Total Amount



Correlation Insights

Insights:

Correlation matrix :

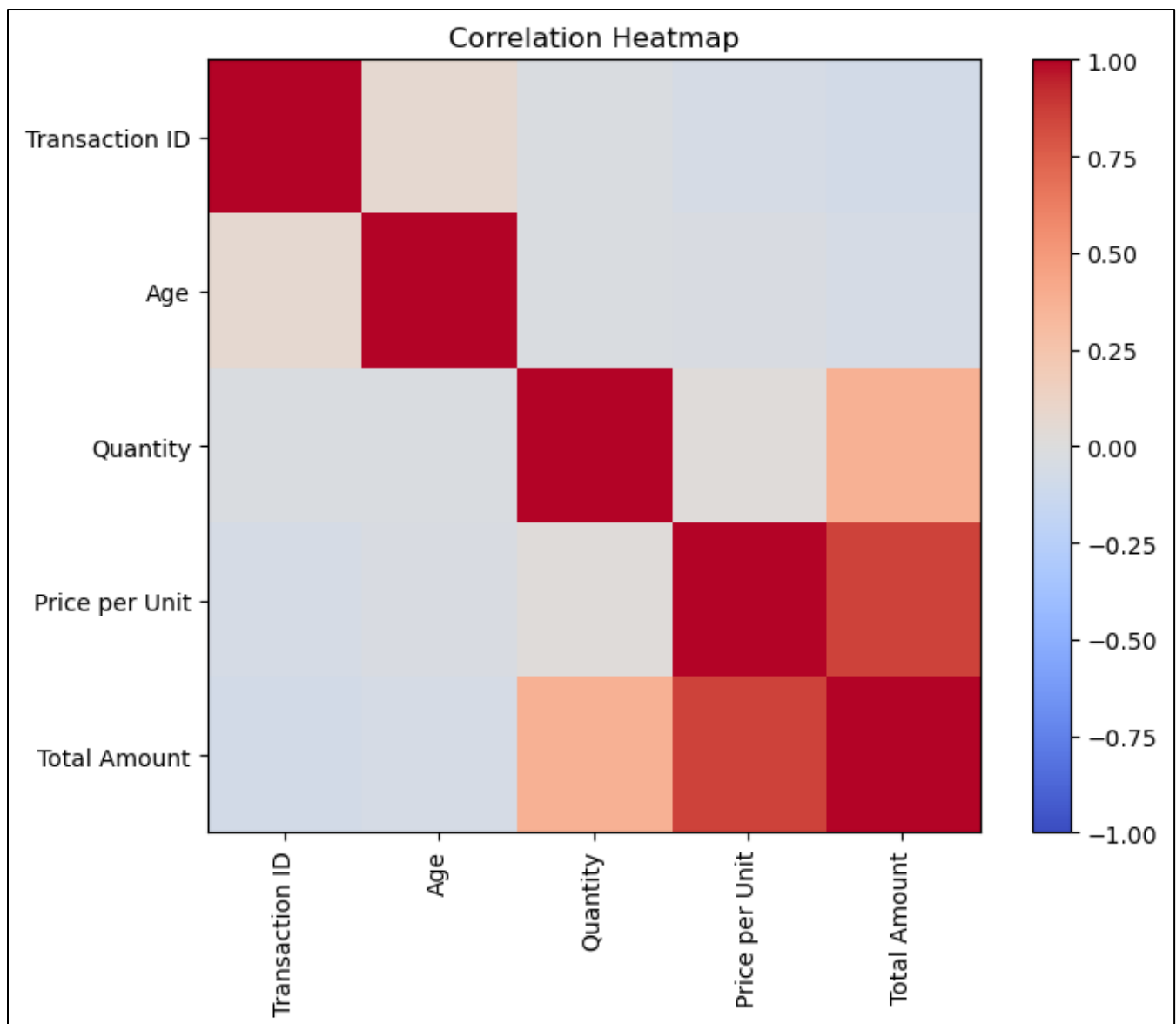
- **Quantity** ↔ **Total Amount** = **0.85** (STRONG positive correlation)
- **Price per Unit** ↔ **Total Amount** = **0.37** (MODERATE positive correlation)
- **Age** ↔ **Total Amount** = **-0.06** (no significant relationship)

This means:

- Higher quantity → higher sales revenue
- Higher price → moderately increases revenue
- Age is NOT a driver of purchase value

Visualizations

Correlation Heatmap



Final Business Recommendations

Recommendations

- Focus marketing efforts during **peak months** (May, Feb, Dec).
- Introduce offers or campaigns in **low-performing period** (September).
- Investigate outlier transactions for:
 - High-margin customers
 - Product bundles
 - Premium pricing opportunities
- Prioritize product categories within mid-price ranges since bulk of purchases fall there.
- Age demographic is stable; targeting based on **product interest** makes more sense than age.